

1. The range of the function $f(x) = \frac{1}{\sqrt{9-x^2}}$ is

- A. $(0, 3)$ B. $[0, 3]$ C. $[\frac{1}{3}, \infty)$ D. $(\frac{1}{3}, \infty)$

2. Given that $f(x) = \frac{1}{x-2}$ and $g(x) = \frac{1}{x}$ then the domain of the function $f \circ g$ is

- A. $\mathbb{R} \setminus \{0\}$ B. $\mathbb{R} \setminus \{\frac{1}{2}\}$ C. $\mathbb{R} \setminus \{0, \frac{1}{2}\}$ D. $\mathbb{R} \setminus \{0, 2\}$

3. (The greatest integer less than or equals x) The range of $f(x) = 4[x]$ is

- A. $\{0, \pm 4, \pm 8, \pm 12, \dots\}$ C. $\{0, \pm 1, \pm 2, \pm 3, \dots\}$
B. $(-\infty, \infty)$ D. $\mathbb{R} \setminus \{0, \pm 1, \pm 2, \pm 3, \dots\}$

4. If the domain of the function $y = f(x)$ is $(4, 7]$ then the domain of $g(x) = f(2-x)$ is

- A. $[4, 7)$ B. $(4, 7]$ C. $(-5, -2]$ D. $[-5, -2)$

5. Given that $f(x) = x^2$ and $g(x) = \begin{cases} 2x & , x \leq 4 \\ x+1 & , x > 4 \end{cases}$ then $(f \circ g)(x) =$

- A. $\begin{cases} 4x^2 & , x < 4 \\ (x+1)^2 & , x \geq 4 \end{cases}$ C. $\begin{cases} 4x^2 & , x < 16 \\ (x+1)^2 & , x \geq 16 \end{cases}$
B. $\begin{cases} 4x^2 & , x \leq 4 \\ (x+1)^2 & , x > 4 \end{cases}$ D. $\begin{cases} 4x^2 & , x \leq 16 \\ (x+1)^2 & , x > 16 \end{cases}$

6. The domain of the function $f(x) = \ln(3 - |9x + 5|)$ is

- a. b. c. d.
A. $\left[-\frac{8}{9}, -\frac{2}{9}\right]$ C. $\mathbb{R} \setminus \left[-\frac{8}{9}, -\frac{2}{9}\right]$
B. $\mathbb{R} \setminus (-\frac{8}{9}, -\frac{2}{9})$ D. $(-\frac{8}{9}, -\frac{2}{9})$

7. The x -intercept of the function $f(x) = -4 + \ln(2x + 3)$ is

- ~~A. $\frac{1}{2}(3 - e^4)$~~ B. $\frac{1}{2}(e^4 - 3)$ C. $\frac{1}{2}e^4$ D. 0

8. The solution set of the inequality $e^{7x} > 2e^{3x}$ is

- ☒ A. $(\ln^4 \sqrt{2}, \infty)$ B. $(-\infty, \ln^4 \sqrt{2})$ C. $(0, \frac{1}{4} \ln 2)$ D. $(-\frac{1}{4} \ln 2, 0)$

9. The range of the function $f(x) = 6 \sin^{-1}(x^2)$ is

- A. $[0, 3\pi]$ B. $[-\frac{\pi}{2}, \frac{\pi}{2}]$ ☒ C. $[-3\pi, 3\pi]$ D. $[-3\pi, 0]$

10. Given that $f(x) = \csc^{-1} x$ then $f(3) =$

- A. $\frac{1}{\sin 3}$ ☒ B. $\sin^{-1}\left(\frac{1}{3}\right)$ C. $\cos^{-1}\left(\frac{1}{3}\right)$ D. $\frac{1}{\cos 3}$

11. If the $f(x) = x^2 + 4x + 7$, $x \in (-\infty, k]$ is one to one function, then the largest possible value of k is

- A. 4 B. 2 ☒ C. -2 D. -4

12. Given that $f(x) = \frac{\pi}{2} + \cot^{-1}(x-2)$ then $f(2) =$

- ☒ A. π B. 2π C. $\frac{5}{2}\pi$ D. undefined

13. The value of $2 \sin^{-1}\left(\frac{\sqrt{7}}{19}\right) + 2 \cos^{-1}\left(\frac{\sqrt{7}}{19}\right) =$

- A. π B. $\frac{1}{2}\pi$ C. $\frac{1}{4}\pi$ ☒ D. 2π

14. $\lim_{x \rightarrow \infty} \left(\frac{\sqrt{25x^2 + x + 1} + 14x + 8}{6x + 4} \right) =$

- ☒ A. $-\frac{5}{6}$ B. $\frac{3}{2}$ C. ∞ D. $-\infty$

15. $\lim_{x \rightarrow 3^-} \left(\frac{1}{x-3} - \frac{27}{x^3-27} \right) =$

- A. -1 B. 1 ☒ C. $\frac{1}{3}$ D. $-\frac{1}{3}$

16. The domain of $f(x) = \cos^{-1}(5x+1)$ is

- A. $[-1, 1]$ B. $[-\frac{1}{5}, 0]$ C. $[0, \frac{2}{5}]$ ☒ D. $[-\frac{2}{5}, 0]$

Name: :-

17. If $y = 0$ and $x = 2$ are horizontal and vertical asymptotes (respectively) of the function

$$f(x) = \frac{ax^3 - x(3x+1)}{x^3 - b} \quad a + b =$$

A. 3

B. 11

C. 8

D. 5

18. $\lim_{x \rightarrow 4^-} \frac{x^2 - 6x + 8}{\sqrt{x^2 - 8x + 16}} =$

A. 22

B. 0

C. does not exist

D. -2

19. The function $f(x) = x^3 \ln|x|$ is.

A. odd

B. neither odd nor even

C. even

D. negative for all $x \in \text{Domain}(f)$

20. If $f(x) = \begin{cases} 3-x, & x < 0 \\ 2+e^{-x}, & x \geq 0 \end{cases}$ then $f^{-1}(x) =$

A. $\begin{cases} x-3, & x > 3 \\ -2+\ln x, & 2 < x \leq 3 \end{cases}$

B. $\begin{cases} 3-x, & x > 3 \\ -2+\ln x, & 2 < x \leq 3 \end{cases}$

C. $\begin{cases} x-3, & x > 3 \\ -\ln(x-2), & 2 < x \leq 3 \end{cases}$

D. $\begin{cases} 3-x, & x > 3 \\ -\ln(x-2), & 2 < x \leq 3 \end{cases}$

CALCULUS I

MID-TERM

20 QUESTIONS, EACH TWO POINTS

1. The range of the function $f(x) = \frac{1}{\sqrt{9-x^2}}$ is

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B. $\mathbb{R} \setminus \left(-\frac{8}{9}, -\frac{2}{9}\right)$ D. $\left(-\frac{8}{9}, -\frac{2}{9}\right)$

7. The x -intercept of the function $f(x) = -4 + \ln(2x+3)$ is

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B. $\begin{cases} 3-x & , x > 3 \\ -2+\ln x & , 2 < x \leq 3 \end{cases}$ D. $\begin{cases} 3-x & , x > 3 \\ -\ln(x-2) & , 2 < x \leq 3 \end{cases}$

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Instructor: د. حبيب ناصر

Date: 6/12/2016.

اسم الطالب: د. حبيب ناصر

Name:

Lecture Time: 12-1

Write the answers

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10
c	a	b	a	b	c	d	d	b	b

Q1) $\lim_{x \rightarrow 3} \ln \sin(e^x - 9) - \ln(x - 3)$

- a) 0 b) $\ln 3$ c) $\ln 6$ d) $-\infty$

Q2) If $f(1) = 3, f'(1) = -3, g(3) = 3, g'(3) = 4, g(1) = -1, g'(1) = 2$, then $\left(\frac{g \circ f}{f}\right)'(1) =$

- a) -3 b) $\frac{6}{9}$ c) $\frac{4}{3}$ d) 3

Q3) If $f(x) = \sec^2 x^3$, then $f'(x) =$

- a) $3\sec x^3 \tan x^3$ b) $6x^2 \sec^2 x^3 \tan x^3$ c) $3x^3 \sec x^3 \tan x^3$ d) $6x \sec^2 x^3 \tan x^3$

Q4) One of the following statements is true

a) If f is continuous at c and g is continuous at c , then $\frac{f}{g}$ is continuous at c

b) If f is differentiable at c , then $\lim_{x \rightarrow c} f(x) = f(c)$

c) If g is continuous at c , then $f \circ g$ is continuous at c

d) If f is continuous at c , then f is differentiable at c

Q5) If $\lim_{x \rightarrow 0} \sqrt{ax^2 + bx} - 3x = 2$, then the values of a and b are

- a) $a = 2, b = 3$ b) $a = 9, b = 12$ c) $a = 2, b = 1$ d) $a = 3, b = 6$

Q6) $\lim_{x \rightarrow 1} \frac{(x-1)}{\sqrt{x^2-2x+1}}$

a) 2

b) -1

☒ c) does not exist

d) 1

Q7) If $f(x) = \begin{cases} cx^2 + 2\cos(x-1) & , x \leq 1 \\ 4x^3 + \frac{3x}{c} & , x > 1 \end{cases}$ is continuous at $x=1$, then $c=$

a) 2, -1

b) -2, 1

c) 1, -3

☒ d) 3, -1

Q8) $\lim_{x \rightarrow 0} 3x \tan\left(\frac{2}{x}\right) =$

a) $\frac{3}{2}$

b) $\frac{2}{3}$

c) 0

☒ d) $\frac{3}{2}$

Q9) $f(x) = \frac{\ln(\cos^{-1} x)}{3x}$

is continuous on $[-1, 1]$

a) $[-1, 0) \cup (0, 1]$

☒ b) $[-1, 0) \cup (0, 1]$

c) $[0, \pi]$

d) $\left(0, \frac{\pi}{2}\right)$

Q10) The horizontal and vertical asymptotes for $f(x) = e^{\left(\frac{x \tan \frac{3x}{2}}{2x}\right)}$

a) $y=e^2, x=1$

☒ b) $y=\sqrt{e}$, No vertical asymptotes

c) $y=\sqrt{e}, x=1$

d) $y=\frac{1}{2}, x=1$

Best wishes

Q 1:

$$\lim_{x \rightarrow 3^+} \ln \sin(x^2 - 9) - \ln(x - 3)$$

$$\lim_{x \rightarrow 3^+} \ln \frac{\sin(x^2 - 9)}{x - 3}$$

$$\ln \left(\lim_{x \rightarrow 3^+} \frac{\sin(x^2 - 9)}{x - 3} \right)$$

$$\ln \left(\lim_{x \rightarrow 3^+} \frac{\sin(x-3)(x+3)}{(x-3)} \right) \Rightarrow y = x - 3$$

When $x \rightarrow 3$
 $y \rightarrow 3 - 3 = 0$
 $y \rightarrow 0$

$$\ln \left(\lim_{y \rightarrow 0} \frac{\sin y}{y} \cdot \lim_{x \rightarrow 3} x + 3 \right)$$

$$\ln(1 \times 6)$$

$$= \ln 6 \Rightarrow \textcircled{C}$$

$$Q 2 :- \frac{f'(1) \cdot (g(f(1)))' \cdot f'(1) - g(f(1)) \cdot f'(1)}{(f(1))^2}$$

$$= \frac{(3 \times 4 \times -3) - (3 \times -3)}{9}$$

$$= \frac{-36 + 9}{9}$$

$$= -3 \Rightarrow \textcircled{A}$$

Q 3 :-

$$f(x) = \sec^2 x^3$$

$$f'(x) = 2 \sec x^3 \cdot \sec x^3 \tan x^3 \cdot 3x^2$$

$$f'(x) = 6x^2 \cdot \sec^2 x^3 \cdot \tan x^3 \Rightarrow \textcircled{b}$$

Q 4:-

(b) If f is differentiable at c , then $\lim_{x \rightarrow c} f(x) = f(c)$

Q 5:-

$$\lim_{x \rightarrow \infty} \sqrt{ax^2 + bx} - 3x = 2$$

* ضرب المرافق

$$\lim_{x \rightarrow \infty} \sqrt{ax^2 + bx} - 3x = \frac{\sqrt{ax^2 + bx} + 3x}{\sqrt{ax^2 + bx} + 3x} = 2$$

$$\lim_{x \rightarrow \infty} \frac{ax^2 + bx - 9x^2}{\sqrt{ax^2 + bx} + 3x} = 2$$

⬅️ طالما أن النهاية جواب (2) $\Rightarrow$ $1x$ درجة البسط = درجة المقام
بالتالي يجب أن تكون قيمة $a=9$

$$\lim_{x \rightarrow \infty} \frac{9x^2 + bx - 9x^2}{\sqrt{9x^2 + bx} + 3x} = 2$$

$$\lim_{x \rightarrow \infty} \frac{bx}{3|x| + 3x}$$

* بما أن $x \rightarrow \infty$
لأخذ أكبر قوة في البسط والمقام.

$$\lim_{x \rightarrow \infty} \frac{bx}{3x + 3x} = 2$$

$$\lim_{x \rightarrow \infty} \frac{bx}{6x} = 2 \Rightarrow \frac{b}{6} = 2$$

$$b = 12$$

So $a = 9$
 $b = 12$

$\Rightarrow$ (b)



Al Balqa' Applied University
Faculty of Engineering Technology
Physics and basic sciences Department

Calculus I first exam

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د. غالب صبر فحوة

Instructor:

Name

Student Number

Duration time:

Date: 1/11/2016

CHOOSE THE CORRECT ANSWER FOR QUESTIONS (1-9) AND COMPLETE THE FOLLOWING TABLE

Q(1)	Q(2)	Q(3)	Q(4)	Q(5)	Q(6)	Q(7)	Q(8)	Q(9)
b	a	d	d	a	d	a	c	b

Q(1) If $f(x) = \frac{\cos x}{|x| - 2}$ then the domain of f is

a) $\mathbb{R} \setminus \{2\}$

c) $\mathbb{R} \setminus [2, 3]$

b) $\mathbb{R} \setminus [2, 3]$

d) $\mathbb{R} \setminus [2, 3]$

Q(2) If $f(x) = 2x + 1$ and $f(g(x)) = x^2 + x + 1$, then $g(1) =$

a) $\frac{3}{2}$

c) 3

b) $\frac{2}{3}$

d) $\frac{-3}{2}$

**For questions (3) and (4): If $f(x) = \ln(2 + \ln x)$, then

Q(3) the domain of f is

a) $(-\infty, e^2)$

c) $(-\infty, e^2]$

b) $[e^{-2}, \infty)$

d) $[e^{-2}, \infty)$

Q(4) $f^{-1}(\ln 2) =$

a) e

c) e^{-2}

b) 2

d) 1

Q(5) If $f(x) = \frac{7}{3 + 2\sin^2 x}$, then the range of f is

a) $\left[1, \frac{7}{3}\right]$

b) $[-1, 7]$

c) $\left[\frac{1}{3}, 1\right]$

d) $[3, 7]$

Q(6) Given that $f(x) = \frac{x^2 + 4}{x^2 - 2x}$ and $g(x) = \frac{1}{x^2 - 1}$, then $\text{Dom}(g \circ f) =$

a) $\mathbb{R} \setminus \{0, 2\}$

b) $\mathbb{R} \setminus \{-2, 0\}$

c) $\mathbb{R} \setminus \{-1, 0, 1\}$

d) $\mathbb{R} \setminus \{-2, 0, 2\}$

Q(7) Let $g(x)$ be the function obtained from $f(x) = [x] + x^2$ by shifting 3 units to the right, 2 units up and reflected about the x -axis, then $g(1) =$

a) -8

b) -2

c) -14

d) -4

Q(8) $\cos^2(\tan^{-1} x) =$

a) $x^2 + 1$

سائق

الحاد

c) $\frac{1}{x^2 + 1}$

b) $\frac{x^2}{x^2 + 1}$

d) $\frac{1}{x^2 + 1}$

Q(9) $\sin(\tan^{-1}(1) + \cos^{-1}(\frac{1}{\sqrt{2}})) =$

a) $1 + \sqrt{3}$

b) $\frac{1 + \sqrt{3}}{2\sqrt{2}}$

c) $\frac{1 + \sqrt{3}}{\sqrt{2}}$

d) $\frac{1 - \sqrt{3}}{2\sqrt{2}}$

Q(10) (Complete the following answer) If $f(x) = \sqrt{(\ln x)^2 - 5 \ln(x) + 4}$, then the domain of f is $(-\infty, e] \cup [e^4, \infty)$

$(-\infty, e] \cup [e^4, \infty)$

$(0, e] \cup [e^4, \infty)$

SPU

$$Q_1 :- f(x) = \frac{\cos x}{[x]-2}$$

$$[x] = \begin{cases} 1 \leq x < 2, & 1 \\ 2 \leq x < 3, & 2 \end{cases} \Rightarrow \text{صنعتی مقام}$$

$$[x]-2=0$$

$$\Rightarrow x=[2, 3)$$

$$\underline{\text{So}} \Rightarrow \left[\mathbb{R} / \frac{1}{2}[2, 3) \right]$$

$$Q_2 :- f(x) = 2x+1 \quad \& \quad f(g(x)) = x^3 + x^2 + x + 1$$

$$x^3 + x^2 + x + 1 = f(g(x)) \Rightarrow \text{بہترین } f(x) \text{ کے لئے } g(x) \text{ کا تعین کیا جائے گا}$$

$$\left(\frac{1}{2}x^3 + \frac{1}{2}x^2 + \frac{1}{2}x \right) \leftarrow g(x)$$

$$\# f(g(x)) = 2\left(\frac{1}{2}x^3 + \frac{1}{2}x^2 + \frac{1}{2}x\right) + 1$$

$$\# f(g(x)) = x^3 + x^2 + x + 1 \Rightarrow g(x) = \frac{1}{2}x^3 + \frac{1}{2}x^2 + \frac{1}{2}x = \frac{2}{2} \Rightarrow \textcircled{a}$$

$$Q_3 :- f(x) = \ln(2 + \ln x)$$

$$\text{Domain} \Rightarrow 2 + \ln x > 0$$

$$\ln x > -2$$

$$\rightarrow \text{بہترین } e \text{ کے طرف سے} \quad e^{\ln x} > e^{-2}$$

$$\boxed{x > e^{-2}}$$

$$\Rightarrow (e^{-2}, \infty) \Rightarrow \textcircled{d}$$

Q 4 :- $f^{-1}(\ln 2) = ?$

⇒ Find $F^{-1}(x)$

⇒ $f(x) = \ln(2 + \ln x)$

$y = \ln(2 + \ln x)$

أذا e الطرفين

$x = \ln(2 + \ln y)$

$e^x = e$

$e^x = 2 + \ln y$

أذا e الطرفين

$\ln y = e^x - 2$

$e = e^{(e^x - 2)}$

$y = e^{(e^x - 2)}$

$f^{-1}(\ln 2) =$

$= e^{(e^{\ln 2} - 2)}$

$= e^{(2 - 2)}$

$= e^0 = 1 \rightarrow \textcircled{d}$

Q 5 :- $f(x) = \frac{7}{5 + 2\sin^3 x}$ Range = ?

① $-1 \leq \sin x \leq 1$

$-1 \leq \sin^3 x \leq 1$

2 × ↙

$-2 \leq 2\sin^3 x \leq 2$

+5 ↙

$3 \leq 5 + \sin^3 x \leq 7$

أخذ العكس ↙

$\frac{1}{3} \geq \frac{1}{5 + 2\sin^3 x} \geq \frac{1}{7}$

7 × ↙

$\frac{7}{3} \geq \frac{7}{5 + 2\sin^3 x} \geq 1 \Rightarrow R: [1, \frac{7}{3}] \Rightarrow \textcircled{a}$

Q 6 :- $(g \circ f)(x) = g(f(x))$

$$g\left(\frac{x^2+4}{x^2-2x}\right) = \frac{1}{\left(\frac{x^2+4}{x^2-2x}\right)^2 - 1}$$

$$= \frac{(x^2-2x)^2}{(x^2+4)^2 - (x^2-2x)^2}$$

(1)

$$= \frac{(x^2-2x)^2}{x^4+8x^2+16 - (x^4-4x^3+4x^2)}$$

$$= \frac{(x^2-2x)^2}{4x^3+4x^2+16}$$

$$4x^3+4x^2+16=0$$

$$x = -2$$

(2) Domain = $D_{\text{result}} \cap D_f \rightarrow$
 $= \mathbb{R}/\{-2\} \cap \mathbb{R}/\{0, 2\}$

$$\frac{x^2+4}{x^2-2x}$$

$$x^2-2x=0$$

$$x(x-2)=0$$

$$\boxed{x=0, x=2}$$

D: $\mathbb{R}/\{-2, 0, 2\} \rightarrow$ (d)

Q 7 :- $f(x) = |x| + x^2$

shifting 3 units to the right $\rightarrow$ تأثير داخلي (طرز انتقال 3)
 $\hookrightarrow |x-3| + (x-3)^2$

2 units up $\rightarrow$ تأثير خارجي (بمقدار 2)
 $\hookrightarrow |x-3| + (x-3)^2 + 2$

reflected about the x-axis $\rightarrow$ (-) * تأثير خارجي

$$\hookrightarrow -f(x) = -|x-3| - (x-3)^2 - 2$$

So $g(1) = -|1-3| - (1-3)^2 - 2 = -8 \Rightarrow$ (a)

Q 8:-

$$\cos^2(\tan^{-1} x) =$$

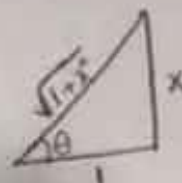
$$\Rightarrow \cos^2(\theta) = \left(\frac{1}{\sqrt{1+x^2}} \right)^2$$

$$= \frac{1}{1+x^2} \Rightarrow \textcircled{c}$$

$$\theta = \tan^{-1} x$$

~ فرض 1

$$\text{So: } \tan \theta = \frac{x}{1}$$



$$R = \sqrt{1+x^2}$$

Q 9:-

$$\sin(\tan^{-1}(1)) + \cos^{-1}\left(\frac{1}{2}\right)$$

$$\sin\left(\frac{\pi}{4}\right) + \frac{\pi}{3}$$

$$= \sin \frac{\pi}{4} \cos \frac{\pi}{3} + \sin \frac{\pi}{3} \cos \frac{\pi}{4}$$

$$= \frac{1}{\sqrt{2}} \cdot \frac{1}{2} + \frac{\sqrt{3}}{2} \cdot \frac{1}{\sqrt{2}}$$

$$= \frac{1+\sqrt{3}}{2\sqrt{2}} \Rightarrow \textcircled{b}$$

$$\tan^{-1} 1 = \frac{\pi}{4}$$

$$\cos^{-1} \frac{1}{2} = \frac{\pi}{3}$$

Q 10:-

$$f(x) = \sqrt{(\ln x)^2 - 5 \ln x + 4}$$

$\Rightarrow$ ~ فرض

$$\ln x = y$$

$$f(x) = \sqrt{y^2 - 5y + 4}$$

$$f(x) = \sqrt{(y-1)(y-4)}$$

$$(y-1)(y-4) > 0$$

$$y-4 > 0$$

$$y > 4$$

$$y-1 > 0$$

$$y > 1$$



$$y > 4$$

$$\ln x > 4$$

$$e^{\ln x} > e^4 \Rightarrow x > e^4 \Rightarrow (e^4, \infty)$$



Faculty of Engineering Technology, Department of Physics and Basic Sciences
Mid Exam, Calculus 101

Instructor name: م. فؤاد

Date: 21-3-2017

number
Zero

$$1 \neq 0 \rightarrow 3x^2 - 27 = 0$$

$$\frac{3x^2}{3} = \frac{27}{3}$$

$$\sqrt{x^2} = |x|$$

~~$k = 0$~~

$$x = \pm 9$$

$$\lim_{x \rightarrow 3^+} \frac{1}{x^2 - 27} = \infty$$

$$\lim_{x \rightarrow 0} \frac{1}{3x^2 - 27}$$

$$\lim_{x \rightarrow 0} \frac{1}{3x^2 - 27}$$

(4)

$$\sin(5x) = 0, x \in [0, \frac{\pi}{5}]$$

$$\sin 5x = 0$$

$$\frac{5x}{3} = \frac{n\pi}{5}$$

$$x = n \frac{\pi}{5}$$

~~Integral~~

$$\Rightarrow x = \frac{\pi}{5}$$

②

Question Three: Find the value(s) of x (if any) that satisfy

$$\ln 1 = 0$$

$$\ln(x-3) = 0$$

$$x-3 = 1$$

$$x = 4$$

$$x = 4$$

Question Four: Evaluate the limit

$$\lim_{x \rightarrow 0} \frac{\tan(3x^6)}{\sin^2(x^3)}$$

$$\lim_{x \rightarrow 0} \frac{\tan(3x^3)}{\sin(x^3)} \cdot \frac{\tan(3x^3)}{\sin x^3}$$

$$\lim_{x \rightarrow 0} \frac{3}{1} \cdot \frac{3}{1} = 9$$

Question Five: Evaluate the limit

$$\lim_{x \rightarrow 8} \frac{\sqrt{x+1}-3}{x-8} \quad \frac{0}{0}$$

$$\lim_{x \rightarrow 8} \frac{\sqrt{x+1}-3}{x-8} \cdot \frac{\sqrt{x+1}+3}{\sqrt{x+1}+3}$$

$$\lim_{x \rightarrow 8} \frac{x+1-9}{(x-8)(\sqrt{x+1}+3)}$$

$$\lim_{x \rightarrow 8} \frac{(x-8)}{(x-8)(\sqrt{x+1}+3)} \Rightarrow \lim_{x \rightarrow 8} \frac{1}{\sqrt{x+1}+3} = \frac{1}{\sqrt{8+1}+3} = \frac{1}{6}$$

Question Six: Evaluate the limit

$$\lim_{x \rightarrow \infty} \tan^{-1} \left(\frac{x^2}{x^2 + 1} \right)$$

$$\frac{\infty}{\infty}$$

$$\lim_{x \rightarrow \infty} \tan^{-1} \left(\frac{1}{1} \right)$$

$$\lim_{x \rightarrow \infty} \tan^{-1}(1) = \frac{\pi}{4}$$

درجه ابطه مساوي
درجه المقام
معامل اقوى حد في البسط
معامل اقوى حد في المقام

Question Seven: Study the continuity of the function $f(x) = \ln(x^4 + 2e^x)$

$$\ln(x^4 + 2e^x)$$

عدد + عدد غير سالب
موجب

الافتراض متحمل على المجال
 $\mathbb{R}$

function is continuity
at $(-\infty, \infty)$

Question Eight: Study the continuity of the following function at $x = 0$

$$f(x) = \begin{cases} 1 - e^{2x} & x > 0 \\ 0 & x \leq 0 \end{cases}$$

$$\lim_{x \rightarrow 0^+} \frac{1 - e^{2x}}{1 - 1} = 0$$

function is continuity
at $x = 0$

$$\lim_{x \rightarrow 0^-} 0 = 0$$

$$\lim_{x \rightarrow 0} 0 = 0$$

21/25 ✓ good.

The Hashemite University
Department of Mathematics

Calculus I
Second Exam

Date 12-11-2012
Time: One Hour

اسم الطالب: د. فلاح محمد الخوالدة الرقم العلمي: ١٥٣١٥٦٧ الرقم المنسل: ٥٥
وقت المعاصرة: 12-1 P.M اسم المدرس: الأستاذ محمد قدومي

1	2	3	4	5	6	7	8	9	10	11	12
a	a	a	a	a	a	a	a	a	a	a	a
b	b	b	b	b	b	b	b	b	b	b	b
c	c	c	c	c	c	c	c	c	c	c	c
d	d	d	d	d	d	d	d	d	d	d	d

1) Let f and g be differentiable at $x = 1$, such that $f(1) = 1$, $f'(1) = 2$, $g(1) = -2$ and $g'(1) = 4$, then $\frac{d}{dx}(g(x)\sqrt{f(x)})|_{x=1} =$

- a) 5 b) 2 c) $-\frac{5}{2}$ d) $\frac{5}{2}$

2) If $f(2) = 1$, $g(x) = \ln(x^2 + 1)$, and $(gof)'(2) = 6$, then $f'(2) =$

- a) 7 b) -3 c) 4 d) 8

3) The equation of the tangent line to the curve

$f(x) = \sin^2 x - x \cos x^2 + 3$ at $x = 0$ is given by

- a) $y = x + 3$ b) $y = x - 3$ c) $y = -x - 3$ d) $y = -x + 3$

4) The critical point(s) of $f(x) = \frac{4}{x}$ is (are)

- a) 0 only b) -2 and 0 and 2 c) -2 and 2 d) 2 only

5) The values of k and m that make

$$f(x) = \begin{cases} \frac{1}{x} + x & , x > 1 \\ kx - m & , x \leq 1 \end{cases} \text{ differentiable at } x=1 \text{ are:}$$

a) $k=0, m=-2$

b) $k=0, m=2$

c) $k=-1, m=2$

d) $k=1, m=2$

6) $\lim_{x \rightarrow \infty} \left(\frac{x+2}{x+7} \right)^{-2x} =$

a) e^4

b) e^6

c) e^8

d) e^{10}

7) The value of (c) such that the line $y = -x - 4$, is tangent to the curve $y = c\sqrt{x}$

a) $c = -4$

b) $c = -2$

c) $c = 2$

d) $c = 4$

8) If $2x^2 + y^2 = 2$, then $\frac{d^2y}{dx^2} =$

a) $-\frac{6}{y^3}$

b) $\frac{4}{y^3}$

c) $-\frac{6}{y^3}$

d) $\frac{6}{y^3}$

9) One of the following is an inflection point for $f(x) = \frac{x^2}{x^2+3}$

a) $(-1, \frac{1}{4})$

b) $(0, \frac{1}{3})$

c) $(1, \frac{4}{7})$

d) $(-2, \frac{4}{7})$

10) $f(x) = \frac{x^2}{x^2+3}$ increasing in

a) $[-1, 1]$

b) $[1, \infty)$

c) $(-\infty, 0]$

d) $[0, \infty)$

11) $f(x) = \frac{x^2}{x^2+3}$ concave down in

a) $[-1, 1]$

b) $(-\infty, -1) \cup (1, \infty)$

c) $(-1, 1)$

d) $[0, \infty)$

12) $f(x) = \frac{x^2}{x^2+3}$ has absolute maximum at

a) $x = -1$

b) $x = 0$

c) $x = 1$

d) non of these

$f(x) = 1/(x+1)$ on $[-\frac{1}{2}, 1]$? If so, find the appropriate value of the number c . (4 points)

Question (8): Sketch the graph of $y = \frac{x^2}{x^2 - x - 2}$ and identify the exact location of all relative extrema, inflection points, where the graph is increasing, decreasing, concave up, concave down, and the asymptotes if any exist. (8 points)

Handwritten notes and a vertical line are visible at the bottom of the page.

(a) $y = \sqrt{13-3x}$

(b) $y = \sqrt{1-3x}$

(c) $y = \sqrt{5-3x}$

(d) $y = \sqrt{9-3x}$

8. The range of the function $\sqrt{4-\sqrt{x}}$ is:

(a) $(0,2)$

(b) $[0,2]$

(c) $[2,\infty)$

(d) $(-\infty,2]$

9. The equation of the line that is perpendicular to the line $x+3y=4$ and its y-intercept 1

(a) $y = 1+3x$

(b) $y = 1-3x$

(c) $y = 1+\frac{x}{3}$

(d) $y = 1-\frac{x}{3}$

10. The amplitude of the function $f(x) = 5-3\sin(2x-\pi)$ is:

(a) 3

(b) -3

(c) 2

(d) 5

11. The function $y = \frac{2x}{x^4+1}$

(a) has a graph that is symmetric about the x-axis.

(b) has a graph that is symmetric about the y-axis.

(c) has a graph that is symmetric about the origin.

(d) is neither even nor odd.

12. The curve with parametric equations $x = \sec t, y = \tan t, -\frac{\pi}{2} < t < \frac{\pi}{2}$ is given by:

(a) $x = \sqrt{1-y^2}$

(b) $x = \sqrt{1+y^2}$

(c) $y = \sqrt{1-x^2}$

(d) $y = \sqrt{1+x^2}$

13. If $f(x) = \frac{1}{\sqrt{x-1}}$, $g(x) = \frac{1}{\sqrt{1-x}}$ then the domain of the function $f \circ g$ is:

(a) $(-1,1)$

(b) $(0,1)$

(c) $(-\infty,1)$

(d) $(1,\infty)$

--	--	--	--	--	--	--	--	--	--

الرقم الجامعي:

اسم مقرر المادة: ...

الرقم المتسلسل: ...

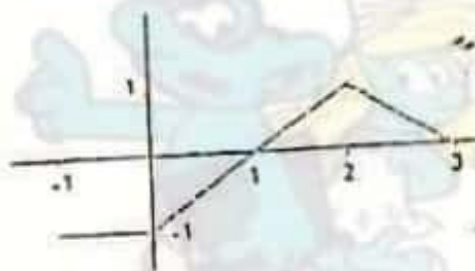
Select the best correct answer and fill it in the following table: (2.5 points each)

	1	2	3	4	5	6	7	8	9	10
(a)										
(b)										
(c)										
(d)										

1. The amplitude and the period for the equation $2y = -4\sin\left(\frac{x}{2} + 2\right)$ respectively, are:

- (a) $-2, 4\pi$ (b) $2, 4\pi$ (c) $-4, 4\pi$ (d) $4, 4\pi$

2. Consider the graph of $y = f(x)$ as shown in the figure

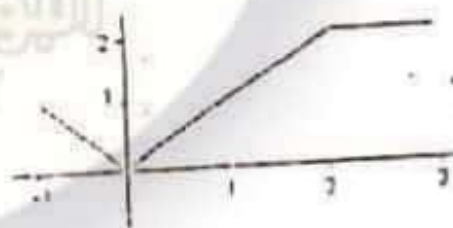


Then the graph of $y = 1 - f(2 - x)$ is:

(a)

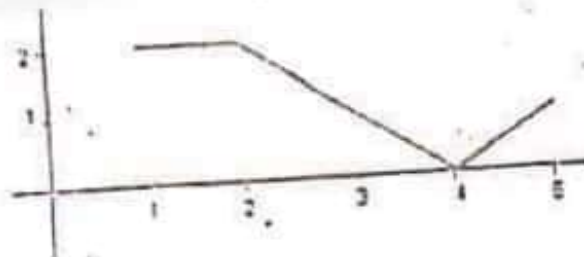
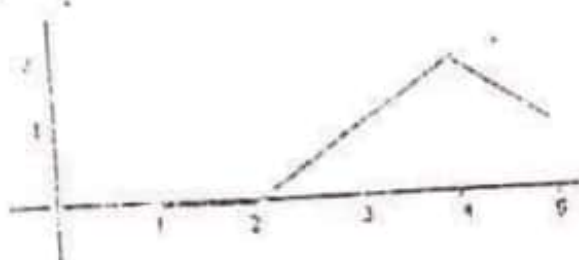


(b)



(c)

(d)



Form D

8. A point is moving along the curve $y = \sqrt{x^3 + 1}$ such that its x-coordinate is increasing at the rate of 2 cm/s, then the distance between that point and the point (2,0) at the instant $x = 2$ is changing at the rate of:

- (a) -2 cm/s (b) 2 cm/s (c) $\frac{8}{3}$ cm/s (d) $-\frac{8}{3}$ cm/s

* Use the function $f(x) = x^{\frac{2}{3}} - x^{\frac{1}{3}}$ to answer the questions 7 - 9:

7. The graph of f has a relative extrema at:

- (a) $x = 0$ and $x = \frac{1}{8}$ (b) $x = \frac{1}{2}$ only (c) $x = 0$ and $x = \frac{1}{2}$ (d) $x = \frac{1}{8}$

8. The graph of f has an inflection point at:

- (a) $x = 1$ only (b) $x = \frac{1}{2}$ only (c) $x = 0$ and $x = 1$ (d) $x = 0$ only

9. At $x = 0$, the graph of f has:

- (a) a vertical tangent line (b) a cusp point
(c) a relative minimum (d) a stationary point

10. Let $f(x) = |\sin x|$. Then: $f'\left(-\frac{\pi}{6}\right) =$

- (a) $\sqrt{3}$ (b) $\frac{1}{2}$ (c) $\frac{\sqrt{3}}{2}$ (d) $-\frac{\sqrt{3}}{2}$

11. Let $y = x^{3/2} - x^{3/4}$. Then, the differential of y at $x = 1$ with $dx = 0.01$ is:

- (a) 0 (b) 0.01 (c) 1.01 (d) $(0.01)^{3/2} - (0.01)^{3/4}$

12. The equation $x^4 + 8 = 10x^2$ has:

- (a) at most one solution in the interval $[-4, 4]$.
(b) exactly one solution in the interval $[-4, 4]$.
(c) at least two solutions in the interval $[-4, 4]$.
(d) no solution in the interval $[-4, 4]$.

8. If $f(x) = |x - 3|$, then $f(x)$ is

- (a) $\left| \frac{x^2 + 7x - 9}{x + 3} \right|$ (b) $\sqrt{\frac{(x-3)^2}{x-3}}$ (c) $\sqrt{\frac{|x-3|^2}{|x-3|}}$ (d) $\sqrt{x^2 - 6x + 9}$

9. $\lim_{x \rightarrow -\infty} x - x^2 + 5x^2 + 11$ is

- (a) 0 (b) $-\infty$ (c) $+\infty$ (d) does not exist

10. If $f(x) = x^2 + 3$, $g(x) = \sqrt{x}$ then the domain of $f \circ g$ is

- (a) $[0, +\infty)$ (b) $(-\infty, 0)$ (c) $(-\infty, +\infty)$ (d) none

11. The graph of $f(x) = 3 - |x - 4|$ is obtained from the graph of $g(x) = |x - 4|$:

- (a) a reflection about the x -axis and a translation 3 units down
(b) a reflection about the y -axis and a translation 3 units up
(c) a reflection about the x -axis and a translation 3 units up
(d) a reflection about the y -axis and a translation 3 units down

12. The domain of $f(x) = \sqrt{2 - \sqrt{x}}$ is

- (a) $[0, \infty)$ (b) $(-\infty, 4]$ (c) $[0, 4]$ (d) $[0, 2]$

13. The graph of $f(x) = x^2 + x$ is obtained from the graph of $g(x) = x^2 - 3x + 1$ by the two translations:

- (a) Two units to the left and one unit up
(b) Two units to the right and one unit down
(c) Two units to the left and one unit down
(d) Two units to the right and one unit up

14. The function $f(x) = \frac{x-5}{|x|-5}$ has removable discontinuity at

- (a) $x = 5, x = -5$ (b) $x = 5$ (c) $x = -5$ (d) none

15. $\lim_{x \rightarrow 2} \frac{\frac{1}{x} - \frac{1}{2}}{\frac{x}{2} - x}$ is

- (a) $+\infty$ (b) $-\frac{1}{4}$ (c) $\frac{1}{4}$ (d) does not exist

16. $\lim_{x \rightarrow \infty} \frac{x^2 - 5}{\sqrt{x}}$ is

- (a) $+\infty$ (b) $-\infty$ (c) 0 (d) does not exist

17. The graph of the curve $x^2 - 5 = y^2$ is symmetric about the

- (a) origin only (b) x -axis only (c) y -axis only (d) origin, x -axis, and y -axis

18. The parametric equations of the circle $x^2 + y^2 = 16$ drawn counter clockwise are

- (a) $x = -4 \cos t$, $y = 4 \sin t$ $0 \leq t \leq 2\pi$
(b) $x = 4 \cos t$, $y = 4 \sin t$ $0 \leq t \leq 2\pi$
(c) $x = 4 \cos t$, $y = -4 \sin t$ $0 \leq t \leq 2\pi$
(d) none

اسم الطالب: الرقم الجامعي: الوقت المتاح:
 الرقم المتسلسل: مدرس المادة:

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
(a)																		
(b)																		
(c)																		
(d)																		

* This exam consists of 18 multiple choice questions, with 1.5 points for each question.

* Select the best correct answer and fill your answer in the above table.

1. The number of vertical asymptotes of $f(x) = \frac{x^3 - 27}{(x-3)(x+4)}$ is
 (a) 1 (b) 3 (c) 2 (d) 0

2. The horizontal asymptote(s) of $f(x) = \frac{x}{|x|-1}$ is(are)
 (a) $y = 1$ (b) $y = -1$ (c) $y = 1, y = -1$ (d) $y = 0$

3. The value(s) of k that makes the function $f(x) = \begin{cases} 3x + 2k^2 & x \geq 0 \\ k & x < 0 \end{cases}$ continuous is(are):
 (a) $\frac{1}{2}$ (b) $0, -\frac{1}{2}$ (c) $0, 2$ (d) 0

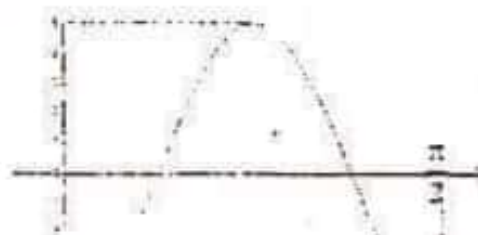
4. If $f(3x+5) = 6x+11$, then $f(x)$ is
 (a) $2x-1$ (b) $2x+1$ (c) $3x+1$ (d) $3x-1$

5. The range of $f(x) = 4 + \sqrt{9-x^2}$ is
 (a) $[4, 7]$ (b) $[0, 4]$ (c) $[4, 6]$ (d) $[-4, 0]$

6. $\lim_{x \rightarrow \infty} \frac{\sqrt{3x^2+11}}{7-4x}$ is
 (a) $-\frac{\sqrt{3}}{4}$ (b) $+\infty$ (c) $\frac{\sqrt{3}}{4}$ (d) does not exist

7. The following graph represent the function

- (a) $f(x) = 5 \cos 2x$
 (b) $f(x) = -5 \cos 4x$
 (c) $f(x) = 5 \cos 2x$
 (d) none



9. $\lim_{x \rightarrow -\infty} \frac{\sqrt{5x^2 - 2}}{x + 2}$

(a) 0

(b) $\sqrt{5}$

(c) $+\infty$

(d) $-\sqrt{5}$

10. The graph of $y^2 = |x| - 5$ is symmetric about the

(a) origin only

(b) x-axis only

(c) y-axis only

(d) origin, x-axis, and y-axis.

11. The vertical asymptote(s) of $f(x) = \frac{2(x^2 + x)}{x^3 - x}$ is(are)

(a) $x = 1$

(b) $x = -1$

(c) $x = -1, x = 1$

(d) $x = 0$

12. $\lim_{x \rightarrow -\infty} \sqrt{25x^2 - 5x - 5x}$ is

(a) $-\frac{1}{2}$

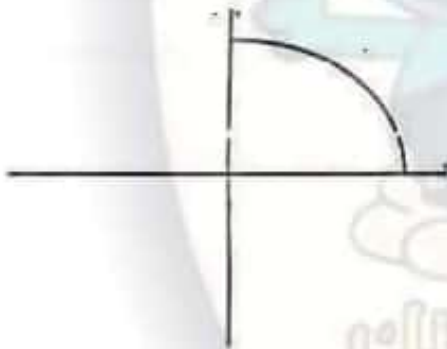
(b) $-\frac{3}{10}$

(c) $+\infty$

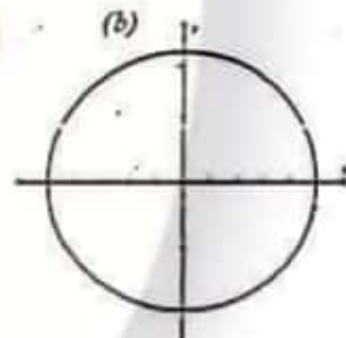
(d) $-\infty$

13. The graph of the parametric curve $x = \sin t$, $y = \cos t$, $-\frac{\pi}{2} \leq t \leq 0$ is

(a)



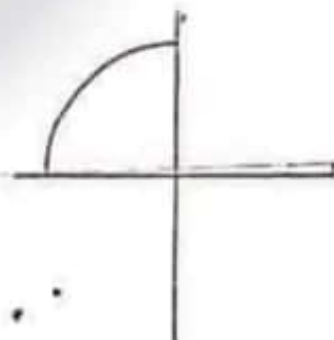
(b)



(c)



(d)



GOOD LUCK

the domain of the function $f(x) = \frac{x}{|x-1|-1}$ is

- (a) $\{0, 1\}$ (b) $\mathbb{R} - \{0, 2\}$
(c) $\{1\}$ (d) $\mathbb{R} - \{2\}$

the range of the function $f(x) = -3\sin^2 x$ is

- (a) $[-3, 0]$ (b) $[-3, 3]$ (c) $[-3, 0]$ (d) $[0, 1]$

the plane curve $y = \sin x \cos y$ is symmetric about the

- (a) x -axis (b) y -axis
(c) origin (d) none of the above

if $f(x) = x - 5x^2$, $x \geq 1$. Then $f^{-1}(x) =$

- (a) $+\sqrt{\frac{1}{100} + \frac{x}{5}}$ (b) $\frac{1}{10} - \sqrt{\frac{1}{100} + \frac{x}{5}}$
(c) $+\sqrt{\frac{1}{100} - \frac{x}{5}}$ (d) $\frac{1}{10} - \sqrt{\frac{1}{100} - \frac{x}{5}}$

the period of the function $f(x) = 3\sin(\frac{x}{2} + 4)$ is

- (a) 4π (b) π (c) $\frac{1}{4\pi}$

if $p > 0$. Then $\lim_{x \rightarrow \infty} x^{-2p+1} = 0$ if

- (a) $\frac{1}{2}$ (b) $p < \frac{1}{2}$ (c) $p < 0$ (d) $p = \frac{1}{2}$

Warning : Calculators are not allowed in this examination.

Part I: Select the best correct answer and fill it in the following table: (2 points each)

	1	2	3	4	5	6	7	8	9	10	11	12	13
a													
b													
c													
d													

1. The graph of the function $f(x) = x^3 - 3x^2 + 3x - 1$ can be obtained by translating the graph of the function $f(x) = x^3$

- (a) left 3 units (b) right 3 units
(c) right 1 unit (d) left 1 unit

2. Let $f(x) = x^3 + 4x + 1$. Then if $f^{-1}(c^3) = c$, then $c =$

- (a) 0 (b) $-\frac{1}{4}$ (c) 1 (d) $\frac{1}{4}$

3. $\lim_{x \rightarrow 3^+} \frac{\sqrt{x^2 - 6x + 9}}{x - 3} =$

- (a) -1 (b) 1 (c) ∞ (d) $-\infty$

4. The function $f(x) = \frac{x^2 - 9}{x^2 - 5x + 6}$ has vertical asymptote(s) at $x =$

- (a) 3 and -3 (b) 3 (c) 3 and 2 (d) 2

5. The function $f(x) = \frac{4x - 9}{\sqrt{x^2 + 5} + 3x}$ has horizontal asymptote(s) at $y =$

- (a) 4 and -4 (b) 4 (c) 1 and 2 (d) 1 and -2

6. Let $f(x) = \frac{x}{x-1}$ and $g(x) = \frac{1}{x-2}$. Then $\text{dom}(f \circ g) =$

- (a) $\mathbb{R} - \{2, 3\}$ (b) $\mathbb{R} - \{2\}$
(c) $\mathbb{R} - \{1, 2, 3\}$ (d) $\mathbb{R} - \{1, 2\}$

7. $\lim_{x \rightarrow 2} \frac{\sqrt[3]{x+6} - 2}{x-2} =$

- (a) $\frac{1}{12}$ (b) $\frac{1}{4}$ (c) $\frac{1}{8}$ (d) $\frac{1}{3}$